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RESEARCH ARTICLE

A Structure-based Data Set of Protein-Peptide Affinities and its Nonredundant Benchmark: Potential Applications in Computational Peptidology

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Abstract: ***Background*:** Peptides play crucial roles in diverse cellular functions and participate in many biological processes by interacting with a variety of proteins, which have also been exploited as a promising class of therapeutic agents to target druggable proteins over the past decades. Understanding the intrinsic association between the structure and affinity ofprotein-peptide interactions (PpIs) should be considerably valuable for the computational peptidology area, such as guiding protein-peptide docking calculations, developing protein-peptide affinity scoring functions, and designing peptide ligands for specific protein receptors.

***Objective*:** We attempted to create a data source for relating PpI structure to affinity.

***Methods*:** By exhaustively surveying the whole protein data bank (PDB) database as well as the ontologically enriched literature information, we manually curated a structure-based data set of protein-peptide affinities, PpI[S/A]$_{DS}$, which assembled over 350 PpI complex samples with both the experimentally measured structure and affinity data. The data set was further reduced to a nonredundant benchmark consisting of 102 culled samples, PpI[S/A]$_{BM}$, which only selected those of structurally reliable, functionally diverse and evolutionarily nonhomologous.

***Results*:** The collected structures were resolved at a high-resolution level with either X-ray crystallography or solution NMR, while the deposited affinities were characterized by dissociation constant, *i.e.* K_d value, which is a direct biophysical measure of the intermolecular interaction strength between protein and peptide, ranging from subnanomolar to millimolar levels. The PpI samples in the set/benchmark were arbitrarily classified into α-helix, partial α-helix, β-sheet formed through binding, β-strand formed through self--folding, mixed, and other irregular ones, totally resulting in six classes according to the secondary structure of their peptide ligands. In addition, we also categorized these PpIs in terms of their biological function and binding behavior.

***Conclusion*:** The PpI[S/A]$_{DS}$ set and PpI[S/A]$_{BM}$ benchmark can be considered a valuable data source in the computational peptidology community, aiming to relate the affinity to structure for PpIs.

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1. INTRODUCTION

Peptides are short strings of amino acids, typically comprising 2-50 residues and normally subdivided into oligopeptides, which contain fewer amino acids (<20), and polypeptides, which possess more amino acids (>20) [1]. In comparison with proteins, peptides are smaller biopolymers and tend to be less structured; they play many crucial roles in regulating the physiological activity and function of other biomolecules and cells [2] and have been exploited as a promising class of biologic drugs to treat a variety of diseases and disorders [3].

In recent years, the notion of peptides has been extended to those functional segments embedded in the intact protein context (*e.g.* short linear motifs [4, 5] and small sequence modules [6]) to fulfill their biological behaviors by mediatingprotein-protein interactions (PPIs), known as peptide-mediated interactions (pMIs) [7]. It is estimated that the pMIs are directly or indirectly responsible for about 15-40% of the cellular interactome, where a short peptide stretch from one protein recognizes and binds to a globular domain of another [8]. Recently, targeting pMIs in omics has gained increased attention in the biology and medicine communities [9] and peptide agents are considered excellent candidates for this therapeutic purpose, such as self-inhibitory peptides [10] and self-binding peptides [11].

Understanding the molecular mechanism of the recognition and interaction between protein receptors and their peptide ligands is fundamentally important for the current computational peptidology [12]. Aprotein-peptide interaction (PpI) contains two different but associated aspects of one biophysical event: how does it interact and how much power? The former and the latter can be characterized by the PpI complex structure and affinity, respectively. Enriched knowledge about the intrinsic structure-affinity relationship of PpIs can considerably facilitate, for example, structure-based peptide design as well as the development and test of new peptide docking [13, 14] and scoring methods [15, 16]. Nowadays, thousands of solved PpI complex structures are available in the protein data bank (PDB) [17], from which a number of secondary databases were deducted to specifically collect, compile, and annotate PpI structural data, such as PepX [18], PeptiDB [19], PixelDB [20] and PepBDB [21]. However, these sources provide neither the PpI affinity data nor the relationship between the structure and affinity of PpI complexes, thus largely limiting their applications in the computational peptidology area. Although few databases such as PDBbind [22] and MOAD [23] as well as compilations [24, 25] provide some PpI affinities, they were not developed specifically for the PpI binding phenomenon and only include limited affinity data. In addition, there are no experimental details available in these sources. Since experimental conditions such as pH, temperature, and ion strength have a significant impact on the measured affinity values, data coming from different sources would result in inevitable errors. Previously, some specific-purpose databases such as AffinDB [26] and SBPPBA [27] were developed for associating structure with an affinity ofprotein-ligand, andprotein-protein complexes, respectively. There should be also an intrinsic relationship between the structural information and binding behavior ofprotein-peptide recognition and association. In this respect, we herein performed an exhaustive survey over the PDB database and ontologically enriched literature information, aiming to develop a structure-based data set of protein-peptide affinities, termed PpI$[S/A]_{DS}$, that collected both the experimentally measured structure and affinity data for various PpIs as well as, more importantly, the association between the structure and affinity of PpIs. In addition, a nonredundant benchmark, termed PpI$[S/A]_{BM}$, was also derived from the set by excluding those informatively redundant samples at sequence, structure, function, and affinity levels. The current set and benchmark consist of 356 and 102 entries, respectively, representing a variety of structurally diverse PpIs as well as their affinities covering a wide range (Fig. **1**).

2. MATERIALS AND METHODS

2.1. PpI Complex Structure Data Recruitment

The structure data of PpI$[S/A]_{DS}$was retrieved from the PDB database [17] by using an advanced search modified from our previous protocol [28], which represents two classes of high-quality structures resolved separately by X-ray crystallography and solution NMR. We adopted the following standards to filter against all the PpI structures deposited in the PDB: (i) high resolution for X-ray structures; (ii) two or more chains; (iii) no DNA or RNA component; (iv) peptide sequence length < 30. Here, only linear peptides were collected.

2.2. PpI Binding Affinity Data Collection

Associating PpI structures to affinities is an error-prone process. Some publications contain incorrect or inconsistent affinity data [29]. The dissociation constant, *i.e.* K_d, is a widely used indicator of the apparent binding strength between a protein receptor and its peptide ligand. This is because K_d can be readily obtained through a number of routine biophysical techniques

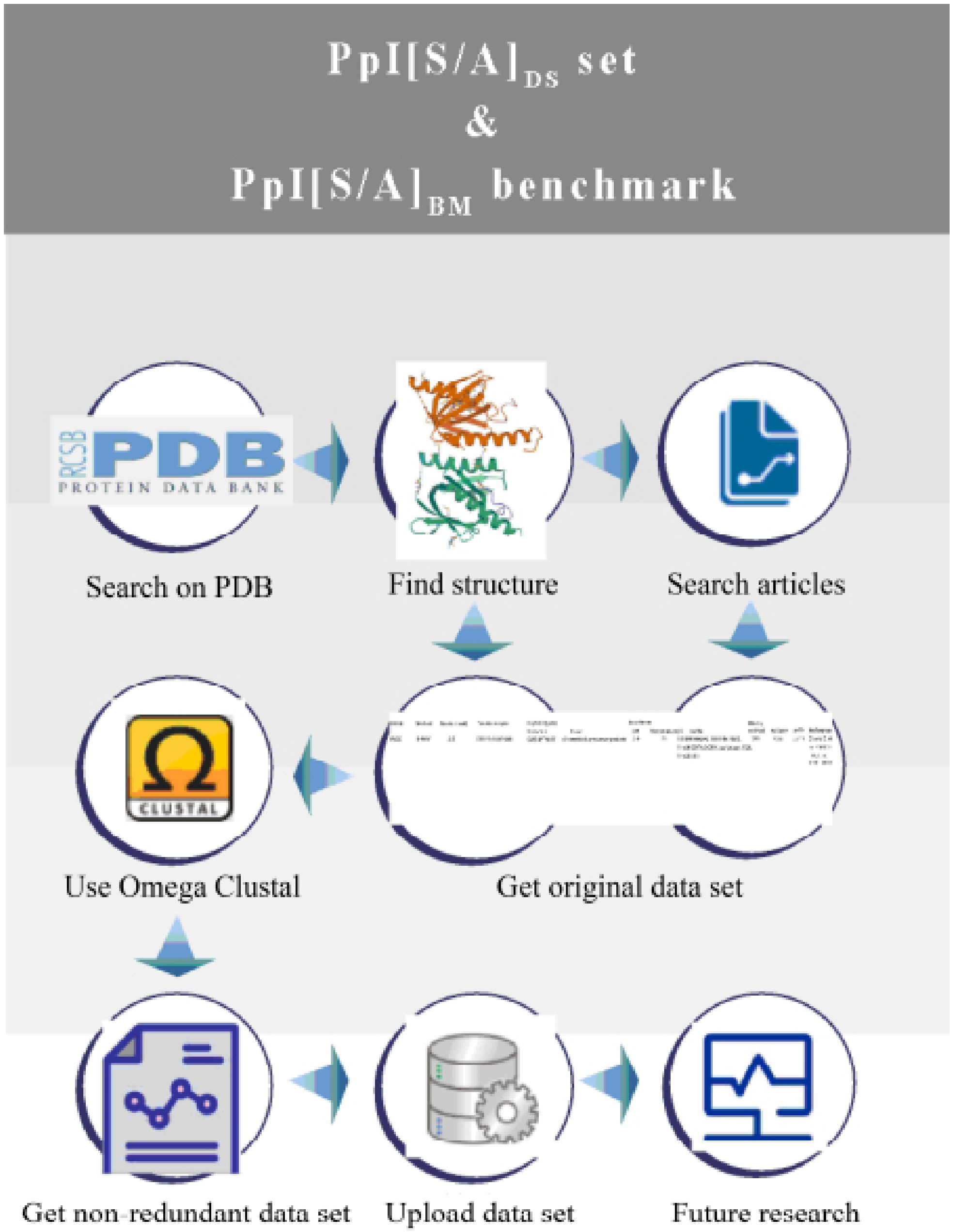

Fig. (1). Overview of generating PpI[S/A]$_{DS}$ set and its nonredundant PpI[S/A]$_{BM}$ benchmark. (*A higher resolution / colour version of this figure is available in the electronic copy of the article*).

such as fluorescence polarization (FP), surface plasmon resonance (SPR), and NMR titration. However, the K_d values are usually reported with standard errors of 20-50% due to measurement error and bias. We found that, for the same complex, there would even be a 3-fold variation in its K_d value if measured with different experimental methods. The discrepancy can be greater when experimental conditions differ; the temperature, ion strength, and pH have a great influence on K_d. Here, the exhaustively surveyed affinity data of the above structure-culled PpI complex samples and the experimental methods used to measure the affinity data are clearly indicated in the literatures. The measured temperature of all affinity data, with the exception of a few below 10 °C and one above 100 °C, were in the range of 10-30 °C, most of which ranged from

18 to 25 °C. The dissociation enthalpy Δ*H*, either determined by isothermal titration calorimetry (ITC) or converted by van't Hoff equation, suggests that the correction does not exceed a factor of 2 in this range. Changes in ion strength and, particularly, pH may also have a significant effect on K_d. For example, if pH changes in a range of 5.5-8.5, K_d can be altered by up to 50-fold.

2.3. Treatment of Redundancy

Peptides usually interact with proteins through conserved residues, also known as short linear motifs (SLiMs) [5]. In order to remove the redundancy of PpI complexes, we first considered the sequence similarity of proteins and peptides. For this, we herein employed the multiple sequence alignment program Clustal Omega to calculate the sequence similarity separately between all proteins and all peptides when making the redundant set [30], and representative PpI complexes were then extracted by using a greedy clustering strategy as a benchmark [31]. In the procedure, all the 356 PpI complexes in PpI[S/A]$_{DS}$ were sorted in decreasing order of protein and peptide sequence lengths in the complexes and a representative list was then created to store selected nonredundant PpI complexes. For each entry in the sorted list, if there is no complex with a similar sequence (identity > 60%) in the representative list, it will be added to the representative list. In this step, a total of 123 nonredundant PpI complexes were obtained and then we manually checked the original data information reported for each sample, from which 21 samples were removed due to serious problems found in the information. Consequently, the resulting PpI[S/A]$_{BM}$ consists of 102 PpIs with their structure-affinity association.

3. RESULTS AND DISCUSSION

3.1. Construction of PpI[S/A]$_{DS}$ Set and PpI[S/A]$_{BM}$ Benchmark

The full list of PpI[S/A]$_{DS}$ set and PpI[S/A]$_{BM}$ benchmark is provided in Supplementary Materials (SM) Tables **S1** and **S2**, respectively; their statistics and content are briefly summarized in Table **1** and schematically organized in Fig. (**2**), respectively. Here, the set and benchmark are separately described below in detail.

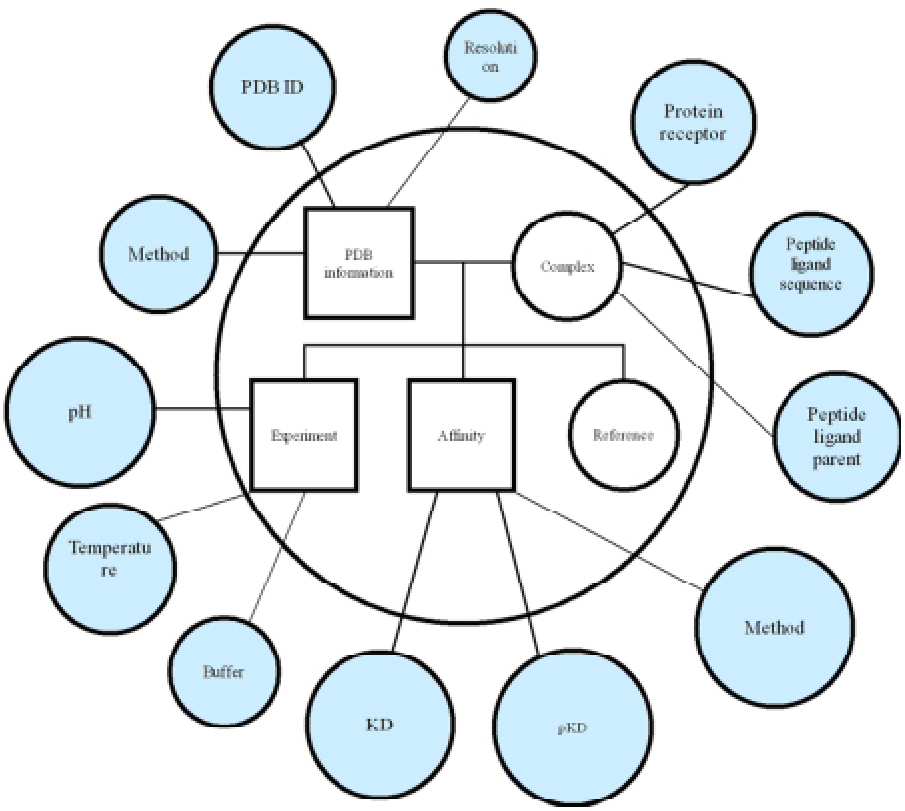

Fig. (2). Schematic representation of the content of PpI[S/A]$_{DS}$ set and PpI[S/A]$_{BM}$ benchmark. (*A higher resolution / colour version of this figure is available in the electronic copy of the article*).

Table 1. Statistics of PpI[S/A]$_{DS}$ set and PpI[S/A]$_{BM}$ benchmark.

Information		PpI[S/A]$_{DS}$	PpI[S/A]$_{BM}$
		356	**102**
Structure	X-ray	182	49
	NMR	174	53
Affinity	SPR	38	14
	ITC	157	50
	FP	115	25
	NMR/others	46	13
Resolution	Range	1.00-3.12 Å	1.14-3.12 Å
	Average	2.05	2.09
K_d	Range	0.00019-2400 μM	0.0006-402 μM
pK_d	Average	5.538	5.698
Sequence length	Range	3-28	4-28
	Average	12.1	15.2

3.1.1. PpI[S/A]$_{DS}$ Set

All PpI complex structure data used in this study were retrieved from the PDB database. There are a total of 356 complex structures in the PpI[S/A]$_{DS}$ set, of which 182 samples were determined by X-ray crystallography, while the other 174 samples were resolved by solution NMR. The resolution of X-ray structures is in the range between 1 and 3.12 Å. The ontological method was used to survey all the literatures associated with these structures through a gene co-citation network [32], which was further one-by-one examined in a manual manner. This is an exhaustive process but may comprehensively create a systematic affinity profile for these structure-resolved PpI complexes from various previous reports. All the curated affinity data were measured by experimental methods, among which 38, 157, and 115 samples were determined by SPR, ITC, and FP, respectively, while the remaining 46 samples were obtained *via* NMR titration, competitive binding assay, or other methods. For a few samples, two or more experimental methods were reported to obtain their affinity data, and there are some inconsistencies and biases between these observations. In fact, the experimental methods used to determine K_d are highly dependent on the protein types and their affinity ranges. Each method has certain limitations and requirements. For example, the SPR, ITC, FP, and some others are applicable for K_d values at nanomolar and micromolar levels. When affinity is below the nanomolar level, the ITC sensitivity is largely reduced and SPR is limited by its reaction kinetics; all these lead to inaccurate or unreliable measurement. In addition, some methods obtained inhibition constant (K_i) for enzyme-peptide inhibitor binding, which needs to be converted to K_d.

3.1.2. PpI[S/A]$_{BM}$ Benchmark

The nonredundant benchmark contains 49 X-rays and 53 NMR structures. The resolution of the X-ray structures ranges from 1.14 to 3.12 Å. For the affinity, 14, 50, 25, and 13 samples were measured by SPR, ITC, FP, and NMR/competitive assay/others, respectively. In the benchmark, about 61% of proteins have moderate size, with lengths ranging from 100 to 400 residues; the other 38 and 2 proteins are less than 100 residues and more than 400 residues, respectively. For peptides in the benchmark, their sequence length is between 4 and 28 amino acids, of which 42% of samples are longer than 15. Here, the distribution of protein and peptide sequence lengths in PpI[S/A]$_{DS}$ set and PpI[S/A]$_{BM}$ benchmark is shown in Fig. (**3**).

3.2. Analysis and Classification of PpI Samples in PpI[S/A]$_{DS}$ Set and PpI[S/A]$_{BM}$ Benchmark

Peptides are highly flexible biomolecules and therefore, in many cases, the peptide ligand structures change significantly upon binding to their protein receptors. The secondary structure of peptides can provide key information as a constraint condition to, for example,protein-peptide docking [33]. In this respect, we herein classified the PpI samples in the PpI[S/A]$_{DS}$ set and PpI[S/A]$_{BM}$ benchmark according to the secondary structure types of peptide ligands based on PDB structural assignment. The peptide secondary structures can be arbitrarily classified into six categories, namely, α-helix, partial α-helix, β-sheet formed through binding, β-strand formed through self-folding, mixed, and other irregular ones.

The classification criteria are as follows:

(i) If half or more peptide residues are helices, it was classified as α-helix (H-class);

(ii) If less than half of peptide residues are helices, it was classified as partial α-helix (pH-class);

(iii) If the peptide is a β-strand adding to the edge of a protein β-sheet, it was classified as a β-sheet formed through binding (bB-class);

(iv) If the peptide is a solitary β-strand when binding to protein, it was classified as a β-strand formed through self-folding (sB-class);

(v) If the peptide is a hybrid of α-helices and β-strands, it was classified as mixed (M-class);

(vi) Others that do not form both α-helices and β-strands were classified as irregular (I-class).

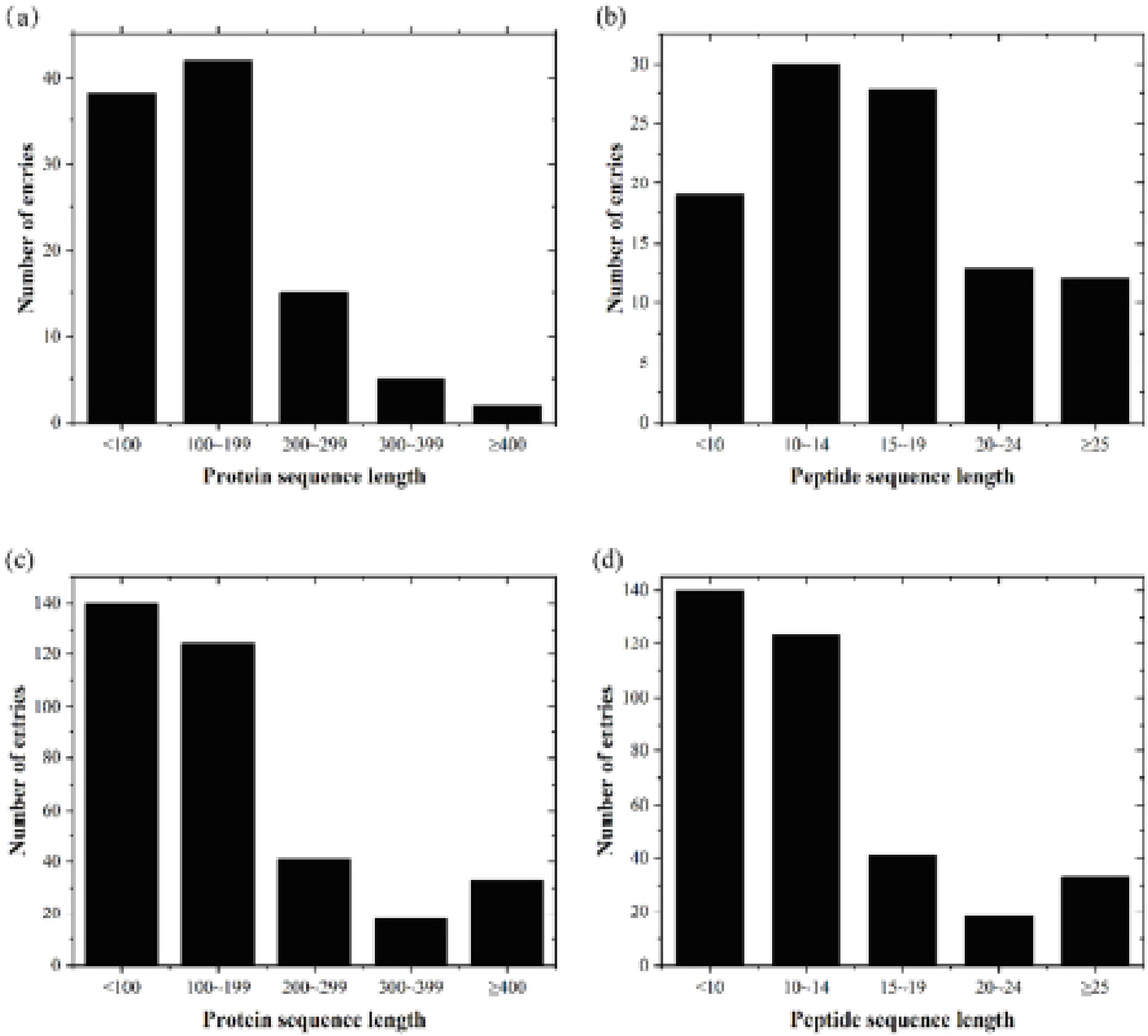

Fig. (3). Distribution of protein and peptide sequence lengths in PpI[S/A]$_{DS}$ set and PpI[S/A]$_{BM}$ benchmark. (*A higher resolution / colour version of this figure is available in the electronic copy of the article*).

Table 2. Summary of PpI structure classification in PpI[S/A]$_{DS}$ set and PpI[S/A]$_{BM}$ benchmark.

Set/benchmark	H-class	pH-class	bB-class	sB-class	M-class	I-class
PpI[S/A]$_{DS}$	54	21	73	4	3	201
PpI[S/A]$_{BM}$	26	10	15	1	1	49

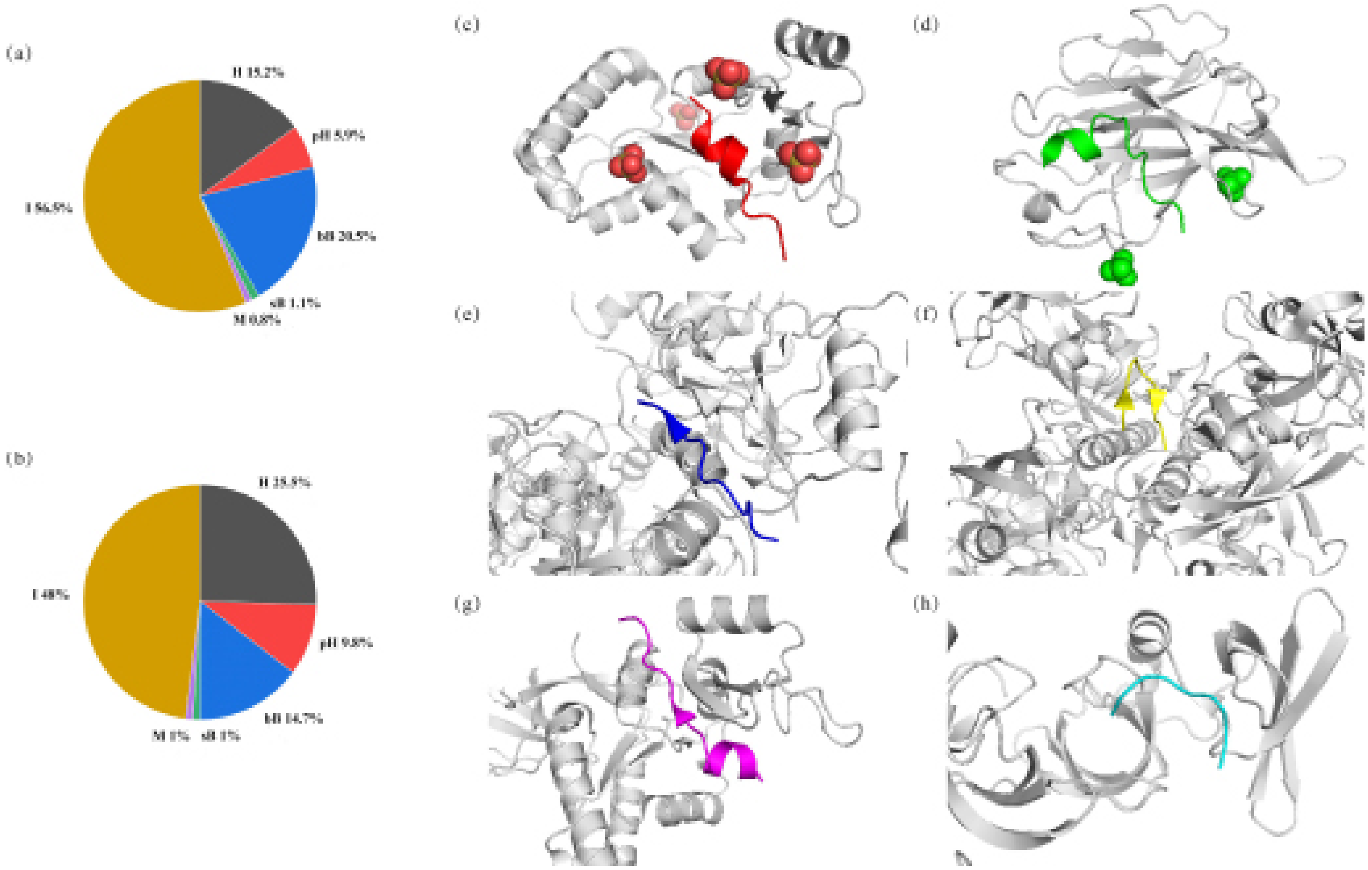

Fig. (4). Distributions of entries in (**a**) PpI[S/A]$_{DS}$ set and (**b**) PpI[S/A]$_{BM}$ benchmark according to the classification of peptide secondary structure types in PpI complexes. An example is provided for each class. Proteins are colored in gray. (**c**) H-class (PDB: 3PW1); (**d**) pH-class (PDB: 2BBA); (**e**) bβ-class (PDB: 3UEO); (**f**) sβ-class (PDB: 3R0H); (**g**) M-class (PDB: 3UAT); (**h**) I-class (PDB: 4CRI). (*A higher resolution / colour version of this figure is available in the electronic copy of the article*).

We did not set coil and other categories in the classification, as such structure classes are lacking in data. The classification results are listed in Table **2** and the distribution of different classes is shown in Fig. (**4**). In the PpI[S/A]$_{DS}$ set, there are 54 and 21, PpI samples belonging to H- and pH-classes, respectively. Although 77 samples involve β-strand or β-sheet structure, only 4 are in the sB-class and the other 73 belong to the bB--class. The M-class contains 3 samples. In addition, the remaining 201 samples are in I-class. For PpI[S/A]$_{BM}$ benchmark, there are 26 in H-class, 10 in pH-class, 1 in sB-class, 15 in bB-class, 1 M-class and 49 in I-class.

While classifying in terms of the secondary structure types of peptide ligands, we also classified these PpI complexes in their biological functions recorded in the PDB database, including enzyme and signaling and regulation. Normally, a PpI system can be regarded as a small peptide ligand bound to a large protein receptor, where the peptide usually serves as a key module to regulate protein function. In this respect, the function classification primarily refers to that of the protein component in complex, which can be divided into 7 classes, namely, enzyme, chaperone, structural protein, signal protein, regulatory protein, peptide-binding protein, and others; some of them also have subclasses. In addition, the PpI binding events can also be classified according to their affinities into high (K_d < 1 μM), medium (1 μM < K_d < 10 μM), moderate (10 μM < K_d < 100 μM), and low (K_d > 100 μM).

As can be seen in Fig. (**5**), class 1 is an enzyme; the complex affinity shows a wide range and is distributed evenly. Class 2 is chaperone; the complex affinity is commonly medium or moderate. Class 3 is a structural protein; the complex affinity is distributed over a wider range. Class 4 is signal protein, which can be further divided into three subclasses as transport, transcription/translation, and other signal proteins. Most of the complex affinities in the class are medium or low, and their distribution is relatively uniform. Class 5 is a

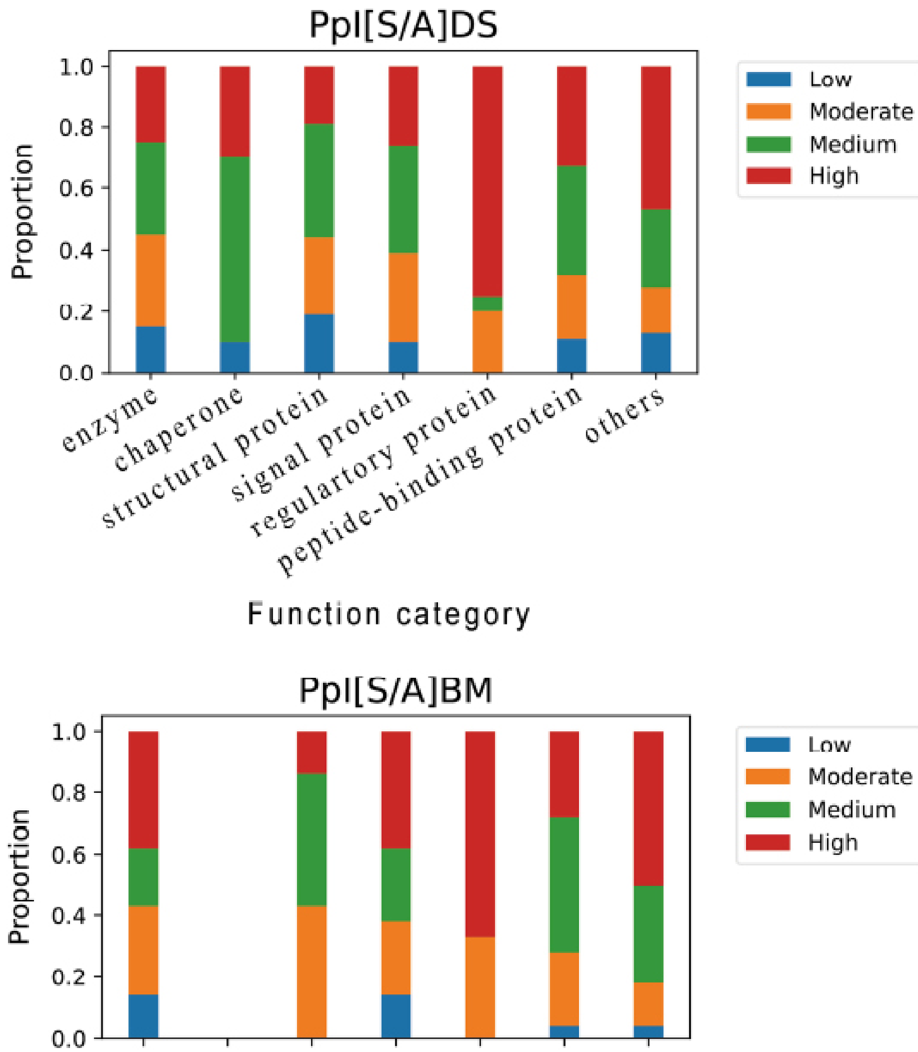

Fig. (5). Histogram representation of PpI function/affinity classification in (a) PpI[S/A]DS set and (b) PpI[S/A]BM benchmark. (*A higher resolution / colour version of this figure is available in the electronic copy of the article*).

regulatory protein, including two subclasses as cell cycle and apoptosis. Only a few samples belong to the class, and the complex affinity is relatively high, most at a high level. Class 6 is a peptide-binding protein, which is generally weak with affinity at low levels. Since the biological functions recorded in the PDB database are confused or inconsistent, the classes of some PpI samples cannot be clearly defined and they are therefore simply put into the class of peptide-binding protein. Few (others) PpI samples exhibit high complexity, which is distributed in a very broad affinity interval and may not have a definite form; they are assigned into class 7, such as toxins, metal binding proteins, viral proteins, *etc.*

Based on published information, the PpIs in the PpI[S/A]$_{BM}$ benchmark become more effective when affinity data are grouped according to the method by which they were determined [27]. Significant correlation can be observed by further dividing the complex affinities into different groups. We found that there was a three-fold difference in affinity for a PpI sample obtained by different experimental methods under the same condition when collecting the data from different sources. Therefore, we herein also grouped the PpI affinities by measure methods to facilitate users (Table **3**).

CONCLUSION

The intrinsic relationship between structure and affinity is always one of the core concerns for the peptidology community considering that the development and test of many new computational methods for treating PpI binding phenomenon essentially rely on the structure-affinity association information. Manual enrichment and careful proofreading of experimental affinity data from various past publications regarding such information is a very exhaustive and trivial task, and we addressed huge effort and spent a long time to do so in this work, which resulted in a PpI[S/A]$_{FL}$ set and a PpI[S/A]$_{BM}$ benchmark for the full collection and nonredundant curation of PpI structure-affinity data, respectively. Moreover, we classified these PpI samples from multiple aspects, such as peptide secondary structures and protein biological functions as well as combined with the affinity levels and measure methods. Since many pH, temperature, and ionic strength have a significant effect on the determined PpI affinity values, all the reported experimental conditions were also compiled in our data. We hope that the classification can be helpful for improving accuracy and reliability when utilizing the set and benchmark to perform further studies. Table **4**.

Table 3. Classification according to the experimental methods used to measure PpI affinities in PpI[S/A]$_{DS}$ set and PpI[S/A]$_{BM}$ benchmark.

PpI[S/A]DS	**High**	**Medium**	**Moderate**	**Low**
SPR	20	7	8	3
ITC	54	62	36	5
FP	42	34	26	13
Others	3	7	15	21
PpI[S/A]BM	**High**	**Medium**	**Moderate**	**Low**
SPR	8	2	4	0
ITC	21	18	9	2
FP	10	8	5	2
Others	0	2	7	4

Table 4. Comparison of PpI[S/A]BM with seven other peptide-protein databases.

Name	**Description**
PepX	A set of structurally diverseprotein-peptide interactions that are clustered based on their binding interfaces, including 505protein-peptide interface clusters from 1431protein-peptide complexes with peptides of less than 10 aa from the PDB.
PeptiDB	A non-redundant database of 103 high-resolution structures of peptide-protein complexes with peptides of 5-15 aa from the PDB.
LEADS-PEP	A representative of the lessons for Efficiency Assessment of Docking and Scoring collection contains 53 protein-peptide complexes with peptide lengths ranging from 3 to 12 residues from the PDB.
PepBDB	A comprehensive and cleaned structural database of biological peptide-mediated interactions with peptide lengths up to 50 aa from the PDB.
PDBbind	A database collects experimentally measured binding affinity data of complexes from PDB. The 2020 version contains binding affinity data for 23,496 complexes, covering protein-ligand, protein-protein, protein-nucleic acid, and nucleic acid-ligand complexes.
PepBank	A database of peptide sequences based on text mining and public peptide data sources with peptides of less than 20 aa.
PepBind	A database of structures, sequences, and experimental observations ofprotein-peptide complexes with peptides of less than 35 aa from the PDB.
PpI[S/A]BM	A set of structurally diverseprotein-peptide interactions including 102protein-peptide complexes with peptide lengths ranging from 4 to 28 residues from the PDB.

ETHICS APPROVAL AND CONSENT TO PARTICIPATE

Not applicable.

HUMAN AND ANIMAL RIGHTS

No humans or animals were used for the studies that are the basis of this research.

CONSENT FOR PUBLICATION

Not applicable.

AVAILABILITY OF DATA AND MATERIALS

The data supporting the findings of the article is available in the Supplementary Materials.

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CONFLICT OF INTEREST

The authors declare no conflict of interest, financial or otherwise.

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Declared none.

SUPPLEMENTARY MATERIAL

The full list of PpI[S/A]$_{FL}$ set and PpI[S/A]$_{BM}$ benchmark can be found in Supplementary Materials Tables **S1** and **S2**, respectively, which are available on the publisher's website along with the published article.

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